

A Data-Driven Automobile Recommendation Model

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Motivation

A recent failure of the transmission on my personal vehicle led me down the exciting, yet daunting path of researching a new vehicle purchase. This vehicle had been in my family since 2014 and was given to me at the age of 16, lasting me well into my early adulthood. Given that the decision to purchase this vehicle was made without my input, I had not done any serious research on the intricacies of a vehicle purchase financially. Even though I was and am still very grateful to be given a car as a teenager, I suddenly and unexpectedly found myself in a situation where I needed to quickly make the biggest financial decision of my life thus far. There is no doubt that millions of people in the United States alone face this issue and do not know where to start.

As I did my research, I visited numerous websites and realized information is so scattered, and for the average car buyer, it is simply too much information to sift through alone. Oftentimes, the buyer will simply show up to a dealership and rely on an interaction with a sales representative to make a major financial decision that may not be in their best interest.

This project aims to develop a simple yet effective data-driven tool to analyze price and reliability across different body types such as sedans, SUVs, and trucks. Users will enter a desired price range and a vehicle body type, and the tool will provide the most reliable vehicles within those constraints, first sorted by reliability then by price.

Data Sources

This project integrates three data sources to analyze vehicles based on price, reliability, and body type. The primary dataset used is a CSV file titled "Cars Datasets 2025.csv" from Kaggle (<https://www.kaggle.com/datasets/abdulmalik1518/cars-datasets-2025>) containing the car manufacturer, model, and pricing data, as the most important factor in a vehicle purchase is often the purchasing price and was the first constraint applied to the integrated dataset used in the model.

In order to further narrow down the search we need to apply a second constraint using the user inputted desired body type using the vPIC API (<https://vpic.nhtsa.dot.gov/api/vehicles>), a public government API. For this purpose, the "Get Models for Make and a combination of Year and Vehicle Type" API method is used to check whether each vehicle within the user specified price range also matches the desired body type using the following format: "/GetModelsForMakeYear/make/{make}/modelyear/2025/vehicletype/truck?format=json." This JSON file returns in the format [{"Make_ID":474,"Make_Name":"HONDA","Model_ID":1866,"Model_Name":"Ridgeline","Vehicle TypeId":3,"VehicleTypeName":"Truck"}].

The third and final data source used is a website called RepairPal (<https://repairpal.com/reliability>), which provides a score out of 5.0 to rate each vehicle model's

reliability, with 5.0 being the highest possible score. Each vehicle model has a separate page, so the reliability scores are scraped using the following URL format: "reliability/{make}/{model}." The request is then parsed by BeautifulSoup and the reliability score is extracted using a regular expression. This score will eventually be used to rank vehicle recommendations.

Integrated Data Model

The Cars Datasets CSV has the following fields and will be the primary data source. Each vehicle make and model includes the engine specs, horsepower, and a handful of other fields, but most importantly the price. The model first removes any dollar signs and blank spaces, then converts the data type from string to integer to clean the data before any price filtering. The price is then used to filter out any models that may be out of the user's specified range.

After the primary dataset is filtered by pricing, the vPIC API is called to filter out any vehicles that do not fit the user specified body type, such as passenger vehicle, multi-purpose passenger vehicle (MPV), or truck. Any specific body types such as coupes or sedans are rolled up into the passenger car category, and any type such as SUV or Minivan is rolled up into multi-purpose passenger vehicles (MPV) to normalize for the API call. The API is called for each make and model year 2025, and each make and model is checked to exist in the API response, and if the make and model does exist, then the data frame is appended with the user specified body type, otherwise it is a null value which are then dropped from the dataset.

Finally, after filtering twice, BeautifulSoup is used to parse the scraped HTML file from the remainder of the makes and model combination pages from RepairPal, and the reliability score assigned to each vehicle is appended to the data frame, with any null values dropped again. Now the final dataset now only includes vehicles with a valid price, body type, and reliability score.

In my original plan, I was going to call this the vPIC API and scrape every single vehicle make and model from RepairPal, and append both to the main dataset, then filter based on the user inputs, but I ran into challenges such as getting blocked by RepairPal, and efficiency issues that caused the program to take longer than necessary to run. Another change is that I had to change the API used, as the original API was not suited for my purpose of finding vehicle body types.

Below is an informal entity relationship diagram to show the integrated data model:

Cars Datasets CSV
Company Names (Make)
Car Names (Model)
Engines
CC/Battery Capacity
Horsepower
Total Speed
Performance (0-100KMPH)
Car Prices
Fuel Types
Seats
Torque

Join on Company Names + Car Names = Make + Model

<u>vPIC API</u>
Make_ID
Make
Model_ID
Model_Name
VehicleTypeID
VehicleTypeName

Join on Make + Model

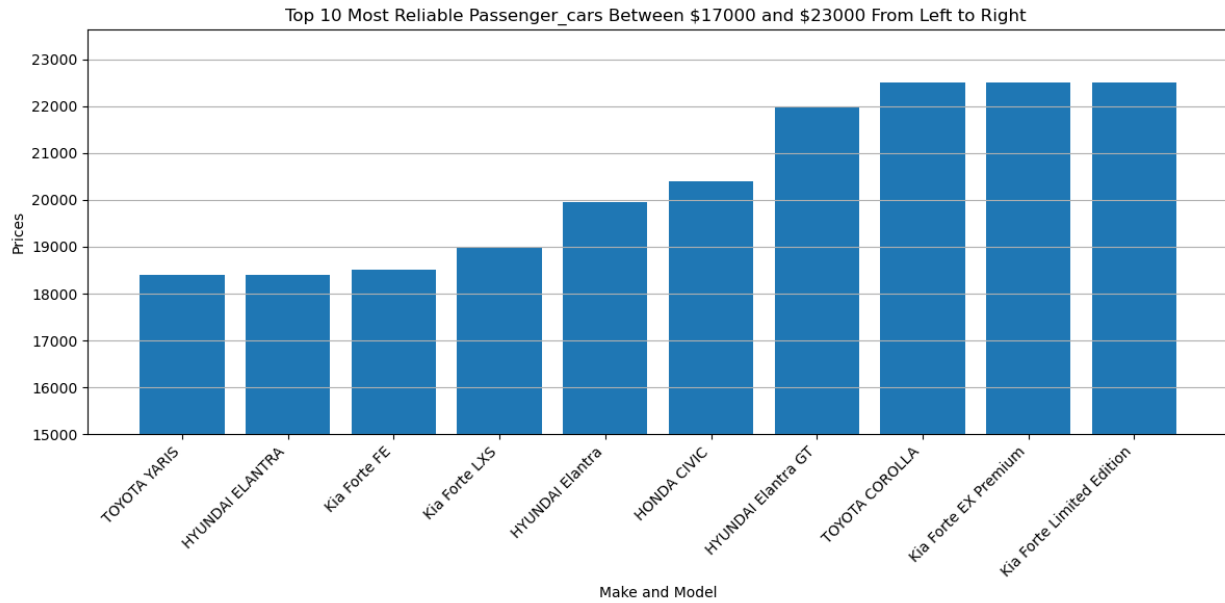
<u>RepairPal HTML</u>
Make
Model
Reliability Score

Analysis/Visualizations

Each make and model is plotted in a bar chart against price, with the most reliable vehicles appearing first, and a price is used as a tiebreaker in the case that the reliability score is the same between multiple vehicles.

The following example shows the output when requesting for recommendations for passenger cars such as sedans and hatchbacks between \$17,000 and \$23,000, using the following arguments in the terminal:

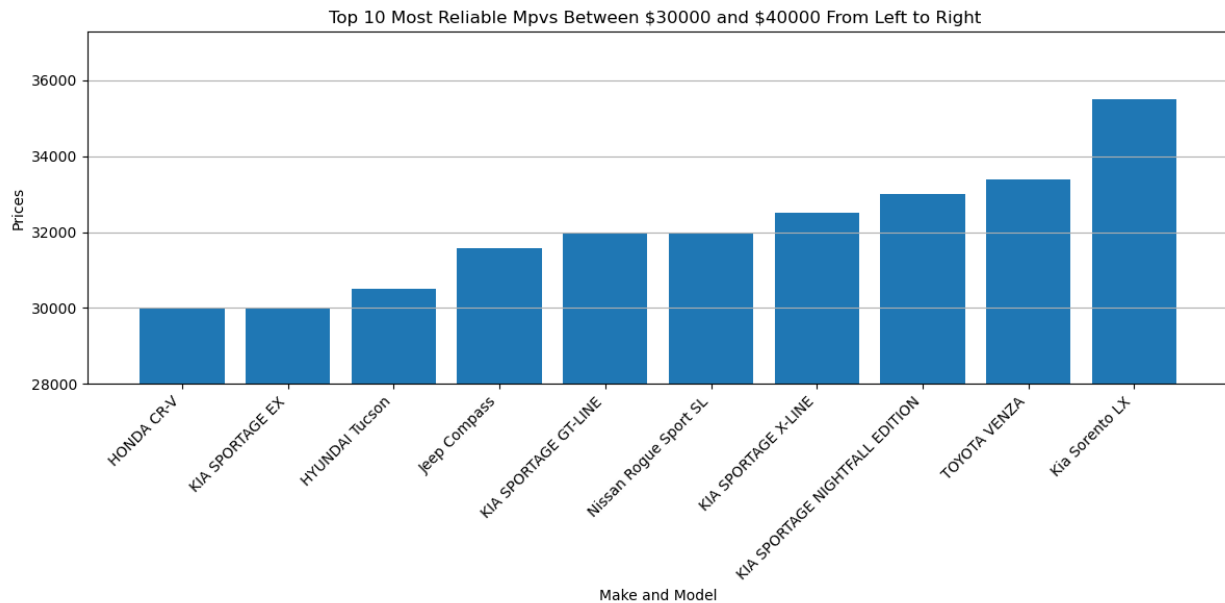
```
--min_price 17000 --max_price 23000 --body passenger_car
```



```
(base) joshuahe@MacBookPro dsci-510-final-project % python src/recommender.py --min_price 17000 --max_price 23000 --body passenger_car
Here Are Your Top 3 Most Reliable passenger_cars:
Cars Prices Company Names Cars Names Reliability Score
0 18400.0 TOYOTA YARIS 4.5
1 18400.0 HYUNDAI ELANTRA 4.5
2 18500.0 Kia Forte FE 4.5
```

The following example shows the output when requesting for recommendations for multi-purpose vehicles such as SUVs or minivans between \$30,000 and \$40,000, using the following arguments in the terminal:

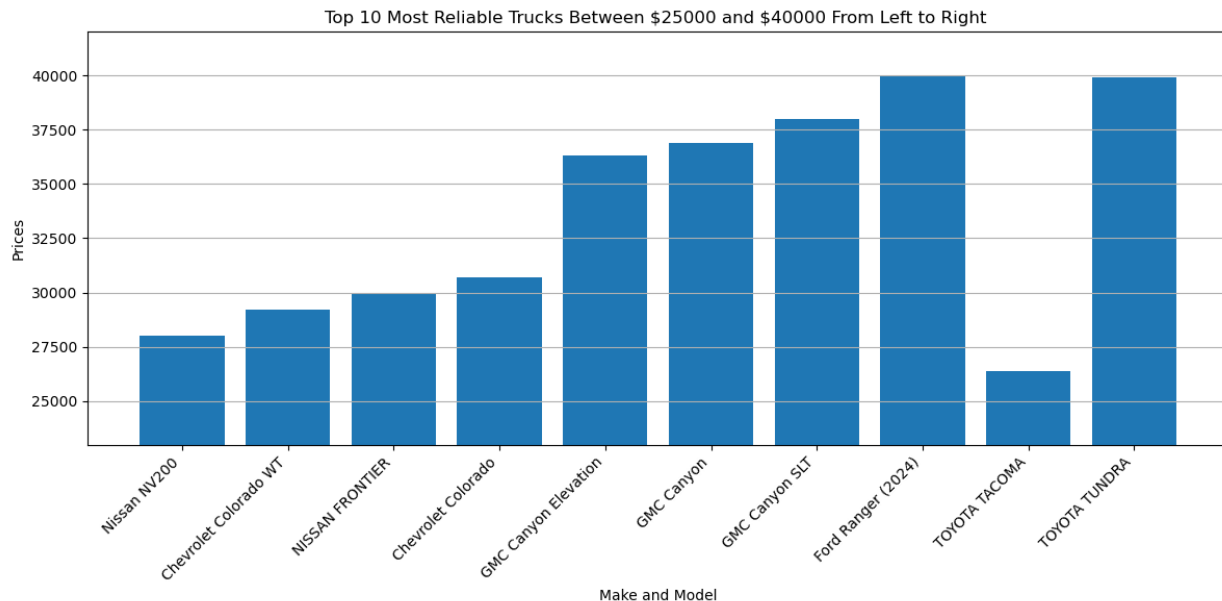
“--min_price 30000 --max_price 40000 --body mpv”



```
(base) joshuahe@MacBookPro dsci-510-final-project % python src/recommender.py --min_price 30000 --max_price 40000 --body mpv
Here Are Your Top 3 Most Reliable mpvs:
Cars Prices Company Names Cars Names Reliability Score
0 30000.0 HONDA CR-V 4.5
1 30000.0 KIA SPORTAGE EX 4.0
2 30500.0 HYUNDAI Tucson 4.0
```

The following example shows the output when requesting for recommendations for trucks between \$25,000 and \$40,000, using the following arguments in the terminal:

`--min_price 25000 --max_price 40000 --body truck`



```
● (base) joshuahe@MacBookPro final_submission % python recommender.py --min_price 25000 --max_price 40000 --body truck
Here Are Your Top 3 Most Reliable trucks:
Cars Prices Company Names Cars Names Reliability Score
0 28000.0 Nissan NV200 4.0
1 29200.0 Chevrolet Colorado WT 4.0
2 30000.0 NISSAN FRONTIER 4.0
```

From the graphs, it can be observed that some vehicles are shown to perform better than their brand reputation would suggest, and others perform worse. This is significant because many people make vehicle purchases based on the perceived brand reputation, rather than doing more detailed research, supporting the need for a data-oriented approach.

Conclusions

This project developed a simple yet effective data-driven tool to analyze price and reliability across different body types such as sedans, SUVs, and trucks. By entering a set of user inputted parameters, the program will give a series of recommendations based on price and reliability, with reliability being the heavier weight in the ranking.

This project also demonstrates the necessity to do data-driven research and analysis when in the process of purchasing a new vehicle. This is due to some vehicles reporting as more reliable than expected, or less reliable than expected. For examples, Toyota has the industry leading brand reputation in terms of reliability, but according to our data model, the Toyota Tacoma is ranked 9th for trucks priced within the \$25,000 to \$40,000 range.

Tying back to my original motivation being that I needed to do my own research on purchasing a new vehicle, I came to the final decision to purchase the Hyundai Elantra, which is also supported by this data model as the 2nd ranked Passenger Car between \$17,000 and \$23,000, especially because it is one of the least expensive options. This shows the usefulness of a data-driven model for identifying vehicles within a budget that is both reliable and financially sound in the long run.

Future Work

If given more time, a few improvements are immediately obvious. First, I would only include one trim of each vehicle, particularly if the same engine and transmission is shared across those trims. Secondly, the engine type, transmission type, and fuel economy/battery capacity are all fields that car buyers are considering beyond just price and reliability, which the current model ignores. Lastly, a cost that buyers often do not consider initially is the insurance cost. Although many degrees more difficult, including an additional estimated insurance premium for each car so that a user could get the full picture of not just the initial purchasing price, but the cost of ownership as well.